

An Improved OPA Model and Blackout Risk Assessment

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Abstract—The ORNL-PSerc-Alaska (OPA) model is a blackout model proposed by researchers at Oak Ridge National Laboratory (ORNL), Power System Engineering Research Center of Wisconsin University (PSerc), and Alaska University (Alaska). Although the OPA model is a landmark study, it has two limitations. First, there is a significant difference between simulation and practice in transmission line outage and update; and second, the simulation of cascading failure and the probability distribution of blackout size are in general not accurate enough. Hence, an improved OPA model is proposed in this paper to address these limitations. The proposed model contains two layers of iteration. The inner iteration describes the fast dynamics of the system and considers the influence of power flow, dispatching, automation, relay protection, and so on. The outer iteration describes the slow overall system evolution and is concerned with the update of the power grid, operation modes and planning. Such a model can be applied to practical large-scale systems. Furthermore, based on the Value at Risk (VaR) and Conditional Value at Risk (CVaR), two new complementary blackout risk indices are defined, which reveal critical characteristics of blackouts and are used to evaluate security levels of power systems. The effectiveness of the improved OPA model is verified by the simulations concerning the Northeast Power Grid of China.

Index Terms—Cascading failure, CVaR, OPA model, VaR.

I. INTRODUCTION

MODERN power systems are becoming highly interconnected. This is especially important in China where fuel energy resources are located in the western inner land and most of the generated electricity is consumed in the major cities in the east coast. Therefore, a nation-wide interconnection has been promoted by the State Power Grid Company. Despite the economic benefits, power system interconnection gives rise to several potential problems, e.g., the high probability of cascading failure. Such problems have drawn special attention after some large-scale blackouts in recent years. In this context, methods based on the complex system theory have become promising in the analysis of power system blackouts [1]–[5].

Several models have been proposed in order to study the cascading failure, such as CASCADE model [6], Branching process model [7], Hidden failure model [8]–[10], OPA model

[11], and the Manchester model [12]. These models capture the salient features of cascading failures from different angles. The CASCADE model and the Branching process model focus on the network characteristics of power systems and as a result are less concerned about other aspects. The Hidden failure model is based on the hidden failure theory and simulates cascading failures after disturbances. The OPA model is built upon the Self-Organized Criticality (SOC) theory, contains two different time scale dynamics, i.e., power flow dynamics and power grid growth dynamics, and reveals the complexity and criticality of power systems. All the models mentioned above are based on dc flows, while the Manchester model is based on ac flows. Similar to the Hidden failure model, the Manchester model is adopted to simulate the cascading failure, but cannot be used to simulate power system evolutions.

Among these models, we focus on the OPA model, which is a novel and powerful tool in studying power system evolution. Paper [13] implements the OPA model in simple networks with regular structures and concludes that a power-dependent tail exists in the probability distribution function (pdf) of the size of blackouts. Paper [14] discovers two critical points, which were further studied in [15]. Motivated by the results in [14] and [15], the authors of [16] carried out the study on the complex dynamics of blackouts. By combining the OPA model with the Manchester model, a new model was proposed in [17] where the dc power flow in the OPA model was replaced by the ac power flow as used in the Manchester model. Later the model in [17] was further developed in [18] to include a voltage stability analysis function. Although models with ac flows are more accurate than the OPA model, they are limited to the study of small-scale systems due to the amount of the calculation involved. In addition, they cannot deal with the shortcomings of the OPA model, among which the most salient one is the critical difference between the simulation and operation in practical systems, such as the outage and update of transmission lines.

In this paper, an improved OPA model is proposed, which considers the effect of dispatching, automation, relay protection and other important elements to make this model more practical than the OPA model. Meanwhile, it continues to use the dc power flow. Therefore, the improved OPA model leads to efficient computation and is applicable to large-scale systems. It is worth mentioning that some steps of this model are inspired by the Hidden failure model.

The rest of this paper is arranged as follows. First, we present the improved OPA model. We investigate in detail the influence of dispatching, automation, communication, relay protection, operation modes and planning. A comparison between simulation results of the OPA model and those of the improved OPA model is also discussed. Second, we introduce two indices, the

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Value at Risk (VaR) and the Conditional Value at Risk (CVaR), which describe the blackout risk quantitatively. Then the cascading failure process in the Northeast Power Grid of China (NPGC) with 570 buses is studied using this improved model. Also by applying this model to the NPGC, we investigate the relationship between power system parameters and blackout sizes, estimate the pdf curves, and calculate the VaR and CVaR indices. Finally, conclusions are drawn in Section VII.

II. SUMMARY OF THE OPA MODEL

It was observed in the OPA model [11] that the slow load increase tends to reduce the margins of transmission lines and hence increases the risk of cascading failure and blackouts. On the other hand, it was assumed that effective operating policies, timely maintenance and well-designed controls, as responses to blackouts, may increase the margins of transmission lines. So the OPA model contains two layers of iterations, the inner iteration is concerned with the fast dynamics which simulate the power flow while the outer iteration cares about the slow dynamics which simulate the growth of power grid such as the increase of generation and load and the update of transmission lines.

As is well known, the OPA model is developed by using the SOC theory. There are two kinds of variables influencing the evolution of power systems. One acts slowly and gently, and the other acts quickly and fiercely. When the system under the influence of these two kinds of variables evolves into a critical condition, a very small disturbance could result in an unpredictable disaster. Scholars have used the OPA model to find out the “critical condition” of power systems. But as discussed below, two parts of the OPA model are inconsistent with practical systems, which undermine the credibility of results derived using the OPA model.

The first part lies in the line outage simulation. In the OPA model, the power flow of every transmission line is bounded by its limit due to dc OPF algorithm. On the other hand, the lines, whose power flows exceed a certain proportion of the limits, are considered overloaded and should be tripped. Such an implementation of tripping lines under limits corresponds to the unwanted operation of relay protection and rarely happens in practice.

The second part lies in the update of the transmission line. Traditional OPA models fail to incorporate the function of planning department thoroughly. The main idea of those models [11], [13]–[16] is to increase the transmission capacity of the lines which are in outage in the inner iteration. In other words, the line will not be updated until they are tripped. Paper [19] tried to strengthen the network by increasing the operating margins of stressed lines according to the current slow dynamics. In practical systems, however, the transmission lines are updated based on the load forecasting, i.e., the future slow dynamics. Therefore, the effect of planning based on load forecasting should be considered in the model.

III. IMPROVED OPA MODEL

A. Introduction of the Improved Model

Aiming to solve the problems in the traditional OPA model as discussed in the last section, an improved OPA model is proposed in this section. This improved model also has a two-loop

structure. The inner loop is the fast dynamics process, which simulates power flow and cascading failure. Compared to the inner iteration of the OPA model, the improved OPA model considers the effects of dispatch, relay protection and automation, etc., and thus reflects the practical systems more faithfully.

The outer one is the slow dynamic process, which simulates the evolution of the power systems. Compared to the outer iteration of the OPA model, the improved model simulates the influence of the effects of planning and operation modes. The time unit of the slow dynamics is usually taken as one day on which the fast dynamics is activated. The process of the improved OPA model can be described as follows.

Step 1) $k = 0$, initialization.

Step 2) The fast dynamics process on day k . Calculate the amount of load shedding.

Step 3) The slow dynamics process. $k = k + 1$. If $k = k_{\max}$, stop the program; otherwise, go to Step 2.

Now we introduce the mathematical description of the model. Suppose there are n buses and m lines. Let p_i^k be the active power injected at bus i on day k , and $p^k = (p_1^k, \dots, p_n^k)^T$ is the vector of active power injections on day k .

The generator set and load set are denoted by G and L respectively. The maximum power limit for generator i on day k is $P_i^{k \max}$, then

$$0 \leq p_i^k \leq P_i^{k \max} \quad i \in G. \quad (1)$$

There is no power loss in dc flow model, so the power balance equation is

$$\sum_{i=1}^n p_i^k = 0. \quad (2)$$

Let F_{jk} be the active power flow on line j on day k . $F^k = (F_1^k, \dots, F_m^k)^T$ is the power flow of m lines on day k . The line flows must satisfy

$$-F_j^{k \max} \leq F_j^k \leq F_j^{k \max}, \quad j = 1, \dots, m \quad (3)$$

where $F_j^{k \max}$ models the transmission limit on line j on day k .

There is a linear relationship in dc flow model between power injections and line flows on day k

$$F^k = A^k p^k \quad (4)$$

where the matrix A^k represents the network constraints and can be calculated easily [8], [11].

Generation and load should be regulated to maintain stability. Notice that, generation re-dispatch is much preferred than load shedding. So p^k should be determined by the following optimization problem:

$$\begin{aligned} \min & \sum_{i \in G} c_i^k p_i^k + \sum_{i \in L} W_i |p_i^k - P_i^k| \\ \text{s.t.} & F^k = A^k p^k \\ & \sum_{i=1}^n p_i^k = 0 \\ & P_i^k \leq p_i^k \leq 0, \quad i \in L \\ & 0 \leq p_i^k \leq P_i^{k \max}, \quad i \in G \\ & -F_{jk}^{\max} \leq F_{jk} \leq F_{jk}^{\max}, \quad j = 1, \dots, m \end{aligned} \quad (5)$$

where P_i^k is the demand power at bus i on day k . Since only p_i^k is supplied, the amount of load $|p_i^k - P_i^k|$ is shed. c_i is the generation cost coefficient for generator i . The coefficient W_i is the economic loss coefficient for the load i , and it is always set as a large number, so that load shedding is much more expensive than generation redispatch.

B. Flowchart of the Improved OPA Model

The flowchart of the improved OPA model is shown in Fig. 1.

C. Fast Dynamics

Since the Hidden failure model represents the cascading failure of power systems more accurately than the OPA model, we can exploit the Hidden failure model in the fast dynamics of the improved OPA model. The fast dynamics is implemented in the following steps.

Step 1) Consider day k . The generator capacity, output of generators and the load are determined by the slow dynamics (see slow dynamics for details). The outage probability of every transmission line is set to be τ to simulate accidental faults.

Step 2) Calculate the dc power flow. The grid may be separated into several islands because of the outage of some lines. Regulation of generation and load is necessary to eliminate the unbalanced power of some islands. The regulation strategy in this paper is as follows. If the total generation is greater than the total load, the output of every generator is decreased proportional to their initial condition in Step 1. If the total generation is less than the total load, the output of every generator is increased according to their reserve capacities. If the total generation capacity is less than the total load, load shedding will be implemented in every load node proportional to its load amount.

Step 3) If there are some lines with flow beyond the limit, then go to step 4. If not, exit the fast dynamics.

Step 4) Calculate the optimization problem (5). In practice, dispatching center may be disrupted due to some contingencies, such as communication interruption and the breakdown of the EMS. In the improved OPA model, the failure rate of dispatching centre is denoted by $1 - \eta$, which is the probability that the optimization program is not calculated. If optimization program is not implemented or convergent, go to Step 5. Otherwise, exit the fast dynamics process.

Step 5) The probability of tripping overloaded lines is set to β (obviously, in this case the probability of failure to operate relay protection is $1 - \beta$). The probability of tripping normal lines is $\xi \times |F/F^{\max}|^n$, where the parameter ξ is the base probability of unwanted operation of relay protection, $|F/F^{\max}|$ is the load rate of the transmission line, and $|F/F^{\max}|^n$ is the $|F/F^{\max}|$ to the power of n . We choose $n = 10$ because the higher the load rate is, the larger the probability of unwanted operation is. If there are outage lines, go back to Step 2; if not, exit the fast dynamics.

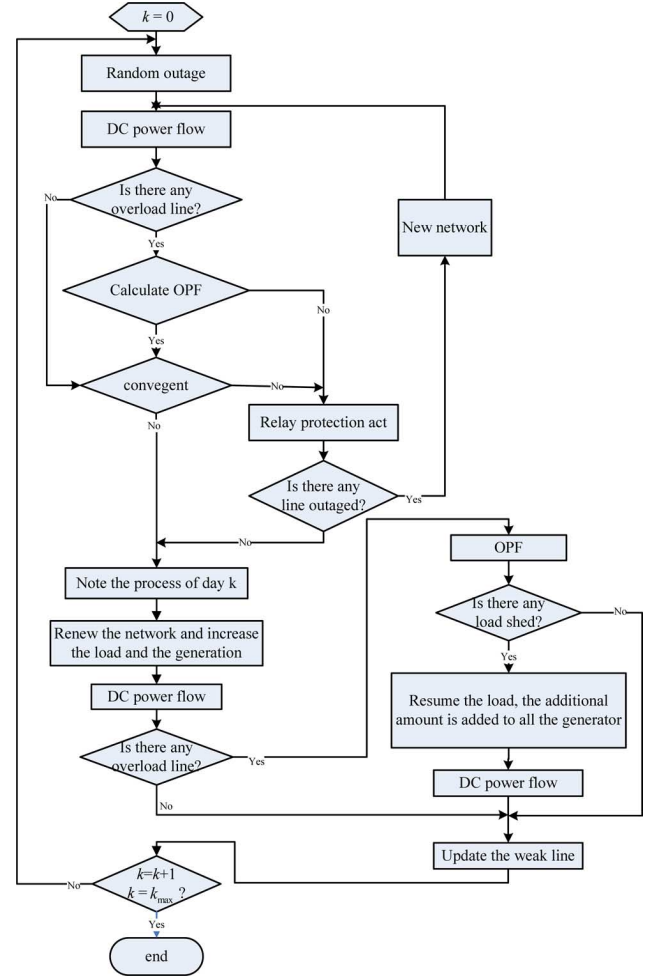


Fig. 1. Flowchart of the improved OPA model.

D. Slow Dynamics

The slow dynamics process of the improved OPA model is as follows.

Step 1) Multiply all loads by a fixed parameter λ which is the average daily rate of increase in electricity demand

$$P_{di}^{k+1} = \lambda P_{di}^k \quad (6)$$

where P_{di}^k is the power demand at Bus i on day k .

Step 2) Increase the generator capacity at the same rate as the power demand

$$P_{gi}^{k+1 \max} = \lambda P_{gi}^k \max \quad (7)$$

where $P_{gi}^k \max$ is the generator capacity at Bus i on day k .

Step 3) According to the power demand and the generator capacity above, calculate the power flow (that is the operation mode of the next day), and update the transmission line which will be overloaded in the next day. Step 3 simulates the effects of the operating and the planning. It is implemented in a sub-program:

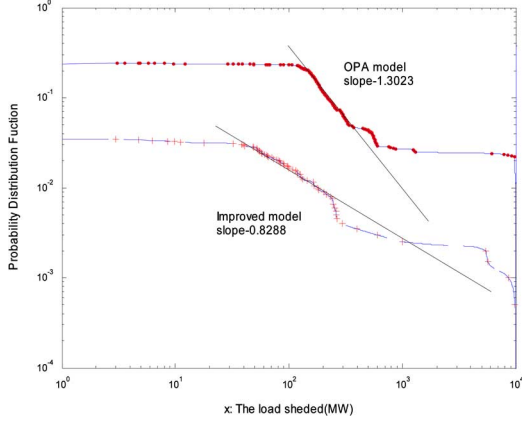


Fig. 2. Pdf of the load shed.

- Calculate the dc power flow. If there are not any overloaded lines, go to Step 3e, otherwise go to Step 3b.
- Calculate the dc OPF (the same as in OPA model). Some transmission lines may not be adequate, and some loads need to be shed. If loads are shed, go to Step 3c, otherwise, go to Step 3e.
- Restore the shed loads. Each generator increases its output proportional to its generation, so the power could be balanced.
- Calculate the dc power flow.
- Rebuild the lines whose $|F/F^{\max}|$ is larger than ε , the load rate of weak lines. This step simulates the department of planning. We update the transmission lines following the OPA model:

$$\begin{aligned} F_j^{k+1 \max} &= \mu F_j^{k \max} \\ Z_j^{k+1} &= \frac{1}{\mu} Z_j^k \end{aligned} \quad (8)$$

where μ is the rate of increase in transmission capacity, Z_j^k is the impedance of the transmission line j on day k .

E. Advantages of Improved OPA Model Over OPA Model

The OPA model and the improved OPA model are applied to the NPGC system to compare the effectiveness of these two models.

The pdf curve of the load shedding amount is shown on a log-log plot in Fig. 2, where the horizontal axis indicates the blackout size and the vertical axis is the corresponding pdf. Notice that we define probability distribution function as

$$P(y) = \int_y^\infty p(x) dx. \quad (9)$$

The procedure of calculating pdf is as follows: First, set the bin width of a continuous variable x . Then, count the number of days that the load shedding amount falls into one bin. The probability is calculated by the corresponding days divided by the total number of days. Finally, pdf is obtained through the

cumulative probability distribution by (9). To draw a continuous pdf curve, piecewise cubic Hermite interpolation is used. The parameters are determined as follows.

- 1) $k_{\max}$
Let $k_{\max} = 2000$, which is more than five years.
- 2) The daily rate of increase in electricity demand
The total load of the NPGC was 18394 MW in July 2005 [20], and 2140 [21] MW in July 2006, so the annual rate of increase in electricity demand is about 16.45%. Then the daily rate of increase in electricity demand is $\lambda = 1.00041$.
- 3) The probability of accident τ
There were 178 accidents in NPGC in 2003 [22]. There are about 700 transmission lines (Here we consider transformers as lines). So the probability of accident is $\tau = 0.0007$.
- 4) The probability of unwanted operation and failure to operate of the relay protection
In practical systems, the probability of unwanted operations and failures of the relay protection are less than 0.01, so set $\beta = 0.999$, and $\xi = 0.001$.
- 5) Other parameters
Let $\eta = 0.95$, namely the failure rate of the dispatching center is assumed to be 5%.
Let $\varepsilon = 0.9$, and this means that the lines will be updated when its flow exceeds 90% of its transmission capacity.
Let $\mu = 1.005$. This parameter must be larger than λ .

From Fig. 2, one could conclude that the number of blackouts by the OPA model is much larger than the one by the improved OPA model. This is because the outage and upgrade criteria of transmission lines are different. In the former 600 days, there are 22 blackouts in the OPA model while 14 in the improved model, mainly due to the first difference. In the latter 1400 days, there are 463 blackouts in the OPA model while 56 in the improved model, mainly due to the second difference.

The improved model is more convincing compared to the OPA model. This is because in reality, no more than ten blackouts happen in this grid within one year, which is much closer to the outcome by the improved model. In addition, the blackout size distribution of the improved model agrees approximately with a power law with exponent of about -0.8288 , and this value agrees with the statistical analysis of the historical data, -0.8641 [5].

IV. VaR AND CVAR

The pdf curve is easy to understand, but it cannot be used to evaluate the blackout risk of the system quantitatively. The VaR and the CVaR are defined in this paper to overcome this shortcoming.

A. VaR

In order to evaluate the investment risk, J. P. Morgan company proposed the VaR when they were developing the Riskmetrics System [23]. Now VaR becomes the most popular theory in the financial analysis. The VaR is the abbreviation of Value-at-Risk that is developed based on the theory of probability statistics. It indicates the largest loss with the confidence level σ in a certain period. The mathematical definition of VaR is

$$\text{Prob}(\Delta V > \text{VaR}_\sigma) = 1 - \sigma \quad (10)$$

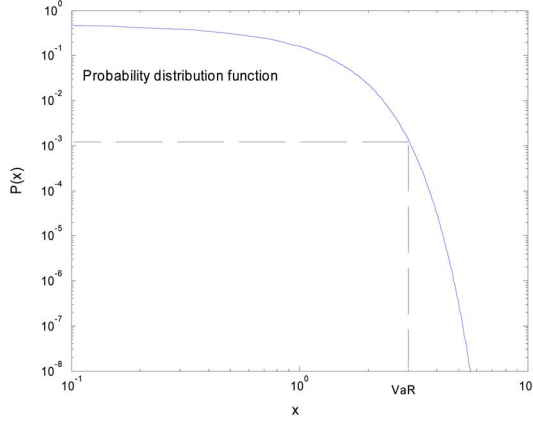


Fig. 3. VaR.

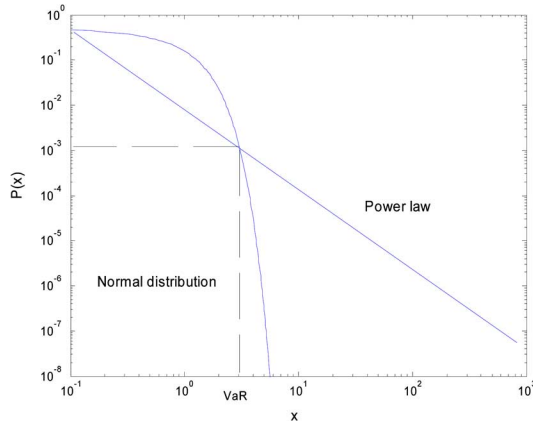


Fig. 4. Long tail.

where σ is the confidence level, ΔV is the loss in a certain period. Here the VaR_σ is computed with the confidence coefficient σ . If the probability density function $p(x)$ is known, the VaR could be calculated by

$$\sigma = \int_{-\infty}^{\text{VaR}_\sigma} p(x) dx. \quad (11)$$

Combining formulation (9), it is easy to see

$$P(\text{VaR}_\sigma) = 1 - \sigma. \quad (12)$$

The sketch of VaR is shown in Fig. 3.

In the figure, y-axis denotes the pdf. The VaR is concerned with not only the size but also the probability. It is easy to understand and calculate, and it can be used to evaluate the risk of different systems. Furthermore, the VaR changes with the confidence level σ .

The VaR has two shortcomings. One is that it does not satisfy the Coherent Axiom [24], [25]; the other is that the tail-loss measurement is not sufficient. It is known that the VaR is a fractile with a certain confidence coefficient, so it is unable to indicate the loss beyond the VaR, especially when the pdf has a fat tail or a long tail (power law). Unfortunately, the long tail is an important characteristic of complex systems in critical conditions and the pdfs of power systems always have long tails as indicated in Fig. 4.

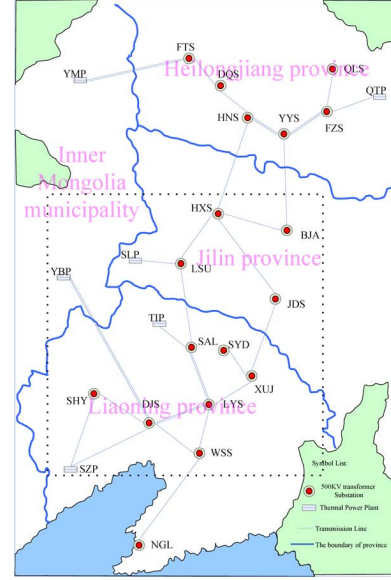


Fig. 5. Northeast power grid of China (500 kV).

Obviously, the risk of the normal distribution is smaller than the risk of the power law distribution. But the VaR at the confidence level σ of these two cases are the same.

B. CVaR

Rockafeller and Uryasev proposed the CVaR to deal with the shortcomings of the VaR [26]. The CVaR is the abbreviation of the Conditional Value-at-Risk, and it is also called Mean Excess Loss, Mean Shortfall, or Tail VaR. It is the conditional mean with a certain confidence coefficient. The mathematical definition of the CVaR is

$$\text{CVaR} = \int_{\text{VaR}_\sigma}^{\infty} xp(x) dx \quad (13)$$

where x is the blackout size and $p(x)$ is probability density function of the blackout size.

The blackout size x and $p(x)$ could be obtained easily based on the improved OPA model. So the VaR and CVaR can be calculated easily.

Take blackout data from North America during 1996–2002 for example, if $\sigma = 0.95$, then $\text{VaR}_{\sigma=0.95} = 5840$ MW, $\text{CVaR}_{\sigma=0.95} = 385$ MW. This implies that there is 95% confident that a cascading failure would not result in a blackout size larger than 5840 MW power of lost. The expected value of blackout size larger than 5840 MW is 385 MW.

V. CASCADING FAILURE IN THE IMPROVED OPA MODEL

From the SOC perspective, when a complex system evolves to a critical condition, the result of disturbances could not be predicted and any small disturbance may result in an unpredictable disaster. We present the progress of a cascading failure which begins with a small disturbance simulated by the improved model. Before that, we introduce the NPGC briefly.

The NPGC consists of four parts, which are Heilongjiang province, Jilin province, Liaoning province and the north of Inner Mongolia municipality. There are about 500 transmission

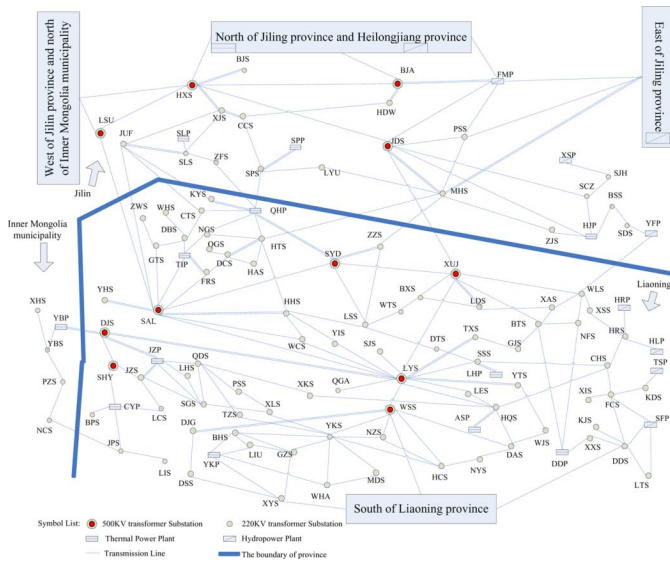


Fig. 6. Northeast power grid of China.

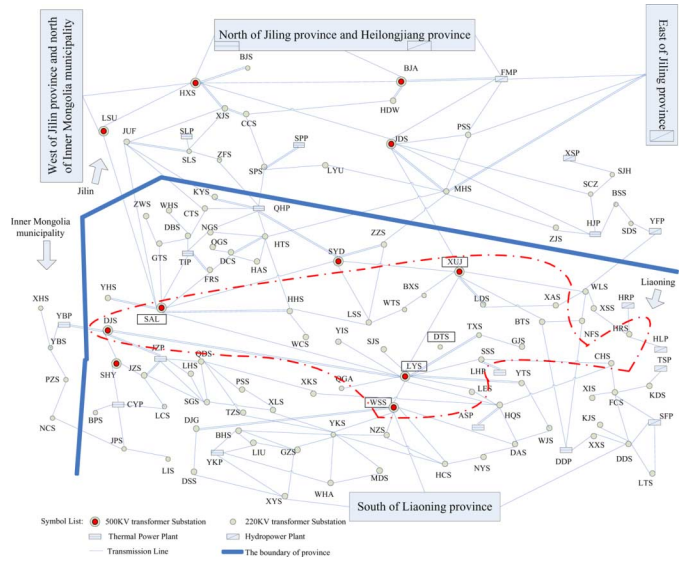


Fig. 8. Stage 2 of the cascading failure.

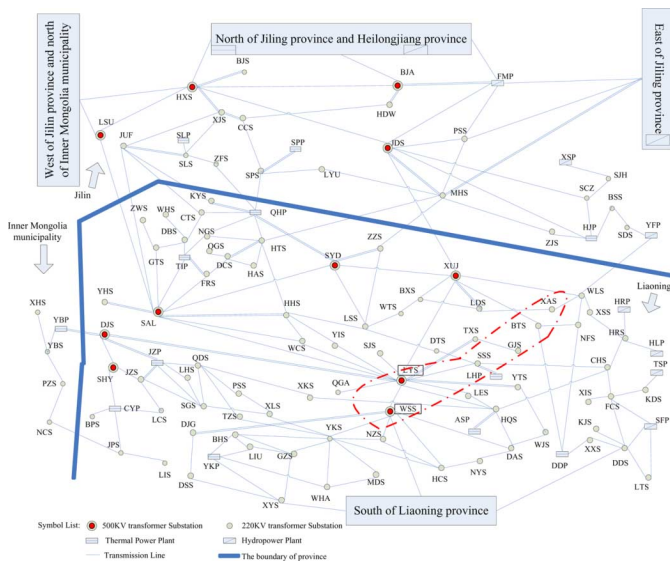


Fig. 7. Stage 0 and Stage 1 of the cascading failure.

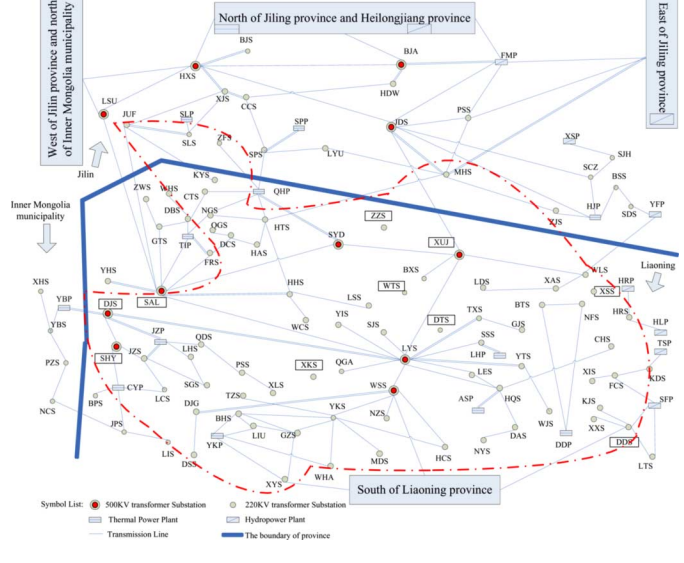


Fig. 9. Stage 3 of the cascading failure.

lines, more than 300 transformers, 250 substations, and about 200 generators. In our study, we consider 500- and 220-kV transmission lines and substations which correspond to a 570-bus system. Fig. 5 is the connection diagram of 500-kV transmission lines and substations.

Fig. 6 is the detailed network connection related to this simulation with irrelevant parts simply represented by panes.

In this system, most of the hydropower plants are located in the east, and most of the thermal power plants are located in the west and Heilongjiang province. The major consumers are in the middle and south of the Liaoning province. Hence, the power is transmitted from the west and the east to the middle and from the north to the south.

The process of a large-scale cascading failure is summarized in Table I. Figs. 7–10 demonstrate the six stages of the cascading failure.

The triggering event of this large-scale cascading failure was an outage transformer of LVS in Liaoning province. The outage of that transformer led to the overload of the neighboring equipments. The dispatching centre didn't notice these problems, due to a certain failure rate of dispatching centre, and thus did not take prompt actions. As a result, all of these equipments are tripped. This led the system to Stage 2, in which substation DTS was separated and all the loads connected to it were shed. In Stage 3, the fault propagates to Jilin province and the system was split into several islands. The Inner Mongolia municipality was involved to this cascading failure in Stage 4. In the last stage, the dispatching centre noticed the situation and responded. After regulating the generations and actively shedding some loads, the cascading failure ended.

This cascading failure began with an outage transformer, and ended with 21.8% loads shed and more than 80 lines (consider transformers as lines) outage. Though it is a simulated cascading

TABLE I
OUTAGED LINES AND TRANSFORMERS AND THE LOADS SHED OF EVERY STEP

Stage	Outage lines	Outage transformers	Loads shed
0		LYS	
1	BTS→GJS, BTS→XAS	LYS, WSS,	
2	LYS→WCS, SAL→WCS, BXS→LDS, CHS→HRS, DTS→LSS, DTS→SSS, HHS→YIS	XJS, SAL	DTS
3	XUJ→BTS, SYD→LSS, CHS→SSS, DAS→WJS, DDP→XXS, DDS→FCS, HCS→NYS, HRS→XSS, JZP→SGS, LSS→WTS, MHS→ZZS, QDS→XKS, QDS→TZS, SGS→XLS, SPS→ZFS, XLS→YKS, YKS→NZS, SYD→ZZS, BXS→WTS, CHS→FCS, CTS→JUF, DAS→WSS, DCS→HTS, DSS→XYS, HCS→NZS, HQS→XKS, KYS→QHP, LSS→ZZS, MDS→WHA, NFS→WLS, QDS→SGS, WLS→XSS	DJS, SHY	LDS, LSS, WLS, XAS, ASP, CHS, DAS, GJS, HQS, LHP, NYS, QGA, SJS, SSS, TXS, WJS, YIS, LES, CTS, DBS DCS, FRS, GTS, HHS, HTS, KYS, NGS, QGS, TIP, WCS, WHA, YHS, ZFS, ZWS, BTS, DDS, FCS, KDS, KJS, LTS, NFS, XIS, XXS, BXS, DJG, DSS, HCS, NZS, SGS, WTS, XKS, ZZS, XSS.
4	YFP→WLS, CYP→LCS, CYP→JPS, DCS→HAS, GZS→XYS, HAS→HTS, HTS→QHP, JPS→NCS, JZS→LCS, NCS→PZS, XYS→YKP	HXS, YBP, ASP, QHP	
5			CHS, DAS, DBS, DCS, FCS, FRS, GJS, GTS, GZS, HAS, HHS, HQS, HTS, KDS, KJS, KYS, LTS, NFS, NGS, NYS, PSS, QGA, QGS, SJS, XLS, XXS, YHS, ZFS, ZWS, YIS, LIU

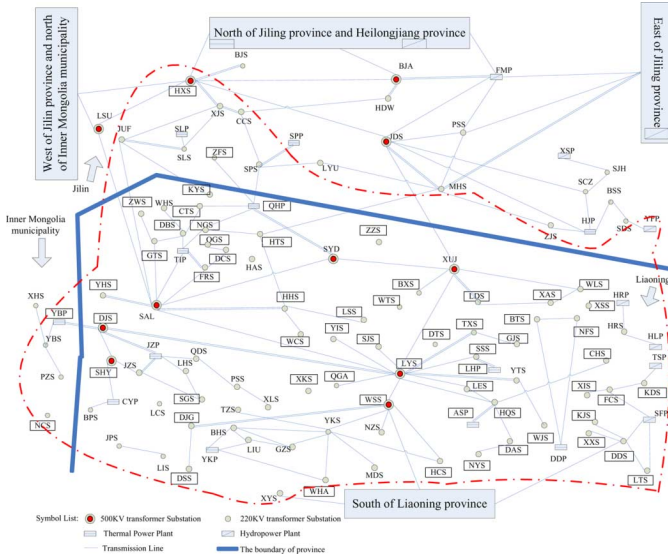


Fig. 10. Stage 4 and Stage 5 of the cascading failure.

failure and did not happen in NPGC, it is possible to happen in practical systems as the North America blackout in 2003. Note that this simulation is similar to the North America blackout to some extent and is helpful for us to further study the cascading failure.

From the above instance, we could see that a small disturbance could cause an unpredictable disaster. That is, in SOC theory, the system has come to the criticality point.

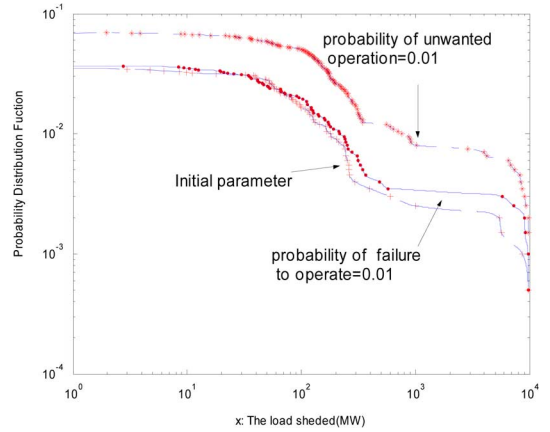


Fig. 11. Pdf curve with different ξ and $(1 - \beta)$.

VI. INFLUENCE OF SOME PARAMETERS

In this section, we investigate the influence of some parameters on the blackout distribution using the VaR and CVaR. The set of parameters in Section III are called initial parameters for convenience.

A. Probability of Unwanted Operation and Failure to Operate of Relay Protection

Three probability distribution curves are shown in Fig. 11. They are with different parameters.

1) The initial parameters;

TABLE II
VaR AND THE CVaR WITH DIFFERENT ξ AND $(1 - \beta)$

ξ	$(1 - \beta)$	VaR	CVaR
0.001	0.001	167.7443	17.1796
0.01	0.001	708.8996	60.1828
0.001	0.01	216.9184	26.8618

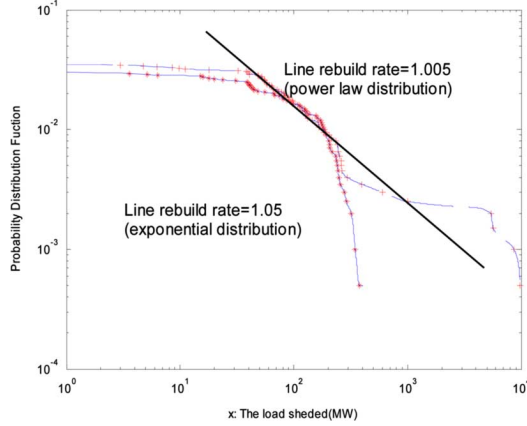


Fig. 12. Pdf curve with different μ .

TABLE III
VaR AND THE CVaR WITH DIFFERENT μ

μ	VaR	CVaR
1.005	167.7443	17.1796
1.05	181.3895	2.4741

- 2) All the other parameters are the same as the initial parameters except the probability of unwanted operation of relay protection(ξ) which is 0.01.
- 3) All the other parameters are the same as the initial parameters except the probability of failure to operate of relay protection($1 - \beta$) which is 0.01.

Let $\sigma = 0.99$, then the VaR and CVaR are listed in Table II.

Table II shows that if ξ and $(1 - \beta)$ become larger, the cascading failure is more serious, and the blackout risk is bigger. Notice that the model could not simulate the short-circuit fault after failure to operate relay protection. So the result underestimates the risk of the failure to operate relay protection.

B. Rate of Increase in Transmission Capacity

Fig. 12 shows two curves with different parameters.

- 1) The initial parameters;
- 2) All the other parameters are the same as the initial parameters except the rate of increase in transmission capacity(μ) which is 1.05.

Let $\sigma = 0.99$, VaR and CVaR are listed in Table III.

Table III shows that increasing μ could reduce blackouts, especially large-scale blackouts. Note that the pdf curve agrees approximately with a power law with μ being 1.005 and that it obeys exponential distribution when μ is 1.05. The system is not in critical condition, and it is safer in SOC theory.

TABLE IV
VaR AND THE CVaR WITH DIFFERENT DISTRIBUTION OF THE LOAD

The distribution of the load	VaR	CVaR
Distribution1	167.7443	17.1796
Distribution2	135.0066	3.4283

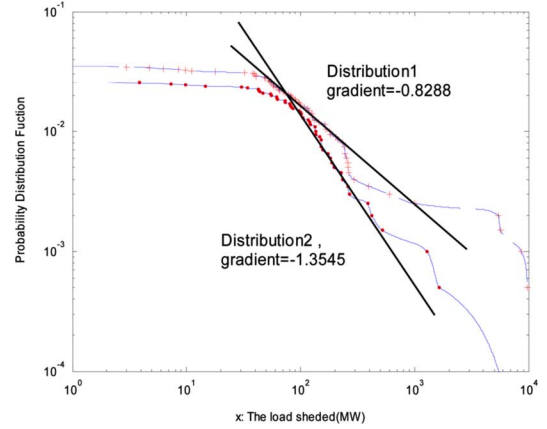


Fig. 13. Pdf curve with different distribution of the load.

After increasing μ to 1.05, the VaR becomes a little larger and the CVaR becomes smaller. This indicates that, increasing μ has no effect on the small-scale blackouts, but could reduce large-scale ones. This can be explained as follows: the reason of small-scale fault is the outage of the transmission line, and the rebuild of the transmission lines has nothing to do with it. The reason of the large-scale fault is the cascading failure, and the rebuild of the transmission line is significant in mitigating the risk.

C. Distribution of Loads

The influence of load distribution is investigated in this section. We consider two distribution strategies.

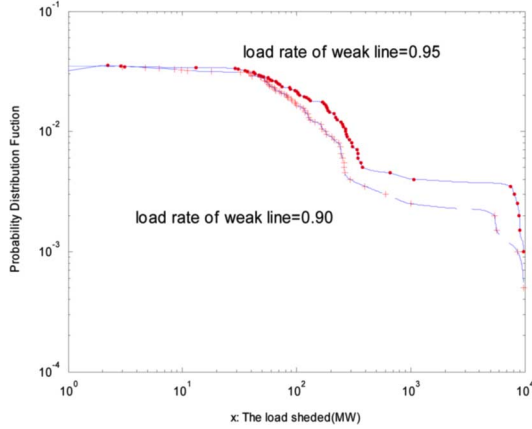
Distribution 1—The real distribution of loads.

Distribution 2—Decrease the load in LiaoNing province, and increase the load in JiLin province and HeiLongJiang province. So the active power is balanced on-site, avoiding transmitting over long distance.

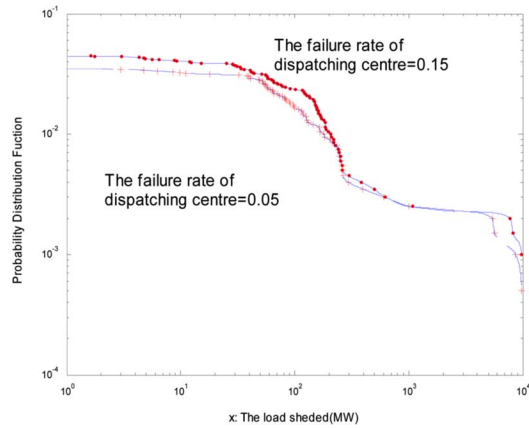
Let $\sigma = 0.99$, the VaR and the CVaR are listed in Table IV.

Table IV shows that the number of blackouts and blackout risk decrease when the active power is balanced on-site. Notice that the blackout distribution is approximately a power law with exponent of about -0.8288 with load distribution 1, and about -1.3545 with load distribution 2. From SOC perspective, distribution 2 is safer than distribution 1 because the smaller the slope is, the less the probability of large-scale blackouts is.

Interconnected power systems make the different regions support each other in case faults take place. However, if vast power is transmitted over long distance, the outage of the tie line is fatal. Once the power is balanced on-site, the tie-line is light-loaded and different regions could support each other in contingencies, in which case the system becomes safer.

Fig. 14. Pdf curve with different ε .TABLE V
VaR AND THE CVaR WITH DIFFERENT ε

ε	VaR	CVaR
0.90	167.7443	17.1796
0.95	268.0006	33.0701

Fig. 15. Pdf curve with different η .

D. Maximum Load Rate of Weak Line

Fig. 14 shows two curves with different parameters:

- the initial parameters;
- all the other parameters are the same as the initial parameters except ε is 0.9.

Let $\sigma = 0.99$, VaR and CVaR are listed in Table V.

The above table shows that the growth in ε could increase the occurrence of large-scale faults, while the total number of blackouts doesn't decrease. This is because if ε becomes larger, the system will not be upgraded in time, and small-scale blackouts may develop into large-scale ones. Hence, the department of planning is important.

E. Failure Rate of Dispatching Centre

Fig. 15 shows two curves with different parameters:

- the initial parameters;

TABLE VI
VaR AND THE CVaR WITH DIFFERENT η

η	VaR	CVaR
0.05	167.7443	17.1796
0.15	197.8015	20.5750

- all the other parameters are the same as the initial parameters except that η is 0.15.

Let $\sigma = 0.99$, the VaR and the CVaR are listed in Table VI.

Table V shows that the dispatching centre is an important department, so the department of automation and communication must provide a reliable environment for dispatchers.

VII. CONCLUSION

This paper summarizes the advantages and disadvantages of the existing blackout models. An improved OPA model is proposed to overcome some disadvantages of the existing models. The proposed model can be applied to large-scale systems, and take into consideration the influence of the effects of dispatching, automation, communication, relay protection, operation mode and planning. Furthermore, based on the VaR and the CVaR, we are able to evaluate blackout risks quantitatively. The VaR and the CVaR are calculated with various choices of parameters, for which appropriate risk mitigation measures are investigated. The effectiveness of the improved OPA model has been verified by the simulation results of the NPGC.

We are currently working on the further development of the proposed model to incorporate more comprehensive descriptions of the modes of important features, e.g., the practical operation of the planning department. While there is still room to improve the proposed model, the theoretical and simulation study in this paper has gained new insight into the mechanism of blackouts in inter-connected large scale power systems.

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